Exponentially Convergent Modular Adaptive Control for Nonlinear Target Tracking

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***Abstract*—A**

***Index Terms*—Neural networks, Adaptive control, Stability of nonlinear systems**

I. Introduction

Depth with nonlinearity creates no bad local minima in ResNets [1].

II. Preliminaries

A. Notation

Let $\mathbb{R}$ denote the real numbers and $\mathbb{Z}$ the integers. For $x \in \mathbb{R}$, let $\mathbb{R}_{\geq x} \triangleq [x, \infty)$, $\mathbb{Z}_{\geq x} \triangleq \mathbb{R}_{\geq x} \cap \mathbb{Z}$, and $\mathbb{R}_{>x} \triangleq (x, \infty)$. Throughout, let $m, n \in \mathbb{Z}_{\geq 1}$ and let $A \in \mathbb{R}^{m \times n}$ be a matrix with columns $\{a_i\}_{i \in [n]} \subsetneq \mathbb{R}^m$. The vectorization of A is denoted by $\operatorname{vec}\{A\} \triangleq \begin{bmatrix} a_1^\top & \cdots & a_n^\top \end{bmatrix}^\top \in \mathbb{R}^{mn}$. The $m \times n$ matrix of zeros is denoted by $0_{m \times n}$. The induced 1- and 2-norms of A are denoted by $\|A\|_1$ and $\|A\|$, respectively. The element-wise matrix of signs of A is denoted by $\operatorname{sgn}(A)$ where $\operatorname{sgn}(0_{m\times n}) = 0_{m \times n}$. The symbols $\subsetneq$ and $\supsetneq$ denote strict subsets and supersets, respectively. For $d \in \mathbb{R}_{>0}$, the open ball centered at $x \in \mathbb{R}^n$ is denoted by $B_d(x) \triangleq \{\zeta \in \mathbb{R}^n : \|x - \zeta\| < d\}$. The corresponding closed ball obtained by taking the closure of $B_d(x)$ is denoted by $\overline{B}_d(x)$. Let $K[\cdot]$ denote the Filippov set-valued map defined in [2, Equation 2b]. Let $\mathcal{A} \subseteq \mathbb{R}^n$ be equipped with Lebesgue measure μ, and let $\mathcal{B} \cong \mathbb{R}^m$ be a finite-dimensional normed space. Define $\|f\|_\infty \triangleq \inf\{M > 0 : \exists N \subseteq \mathcal{A}$ with $\mu(N) = 0$ and $\|f(x)\| \leq M$ for all $x \in \mathcal{A} \backslash N\}$. Let $\mathcal{L}_\infty(\mathcal{A}, \mathcal{B})$ be the collection of all measurable $f : \mathcal{A} \to \mathcal{B}$ for which $\|f\|_\infty < \infty$. A continuous function $f : \mathcal{D} \to \mathcal{R}$ is said to be of class $\mathcal{C}^k(\mathcal{D}; \mathcal{R})$ if it has $k \in [0, \infty] \cap \mathbb{Z}$ continuous derivatives, where the domain $\mathcal{D} \subseteq \mathbb{R}^n$ and codomain $\mathcal{R} \subseteq \mathbb{R}^m$ are occasionally dropped for brevity. A function $\rho : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ is said to be of class $\mathcal{P}_\infty$ if it is of class $\mathcal{C}^0$, strictly increasing, and possesses the property that $\lim_{s \to \infty} \rho(s) = \infty$. To denote a positive definite matrix, $A \succ 0$ is written.

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B. Deep Residual Neural Network Model

The residual neural network (ResNet) architecture, characterized by skip connections and hierarchical feature extraction, models incremental changes rather than complete transformations of the underlying nonlinear mapping. This architecture learns the differences (or "residuals") between the input and desired output at each layer, thereby enabling effective function approximation for complex nonlinear systems without requiring explicit governing equations. A fully-connected feedforward ResNet is constructed as follows.

Let $b \in \mathbb{Z}_{\geq 0}$ be the number of residual blocks, let the input be $x \in \mathbb{R}^{L_{\text{in}}}$, and let the output be $y \in \mathbb{R}^{L_{\text{out}}}$, where $L_{\text{in}}, L_{\text{out}} \in \mathbb{Z}_{\geq 1}$. For each block index $i \in \{0, \ldots, b\}$, let $k_i \in \mathbb{Z}_{\geq 1}$ denote the number of hidden layers in the i^{th} block, let $\kappa_i \in \mathbb{R}^{L_{i,0}}$ denote the block input, and let $\theta_i \in \mathbb{R}^{p_i}$ denote the vector of parameters (weights and biases) associated with the i^{th} block. Note that $\kappa_0 \triangleq x$ and $L_{0,0} \triangleq L_{\text{in}}$.

For all $(i, j) \in \{0, \ldots, b\} \times \{0, \ldots, k_i + 1\}$, let $L_{i,j} \in \mathbb{Z}_{\geq 1}$ denote the number of neurons in the j^{th} layer for the i^{th} block. For all $(i, j) \in \{0, \ldots, b\} \times \{0, \ldots, k_i\}$, define the augmented dimension $L_{i,j}^a \triangleq L_{i,j} + 1$. Each block function $\Phi_i : \mathbb{R}^{L_{i,0}^a} \times \mathbb{R}^{p_i} \to \mathbb{R}^{L_{i,k_i+1}}$ is a fully connected, feedforward deep neural network (DNN) with $L_{i,k_i+1} \triangleq L_{\text{out}}$ for all $i \in \{0, \ldots, b\}$. For any input $v \in \mathbb{R}^{L_{i,j}^a}$, the DNN is defined recursively by

$$\varphi_{i,j}(v) \triangleq \begin{cases} V_{i,0}^\top v, & j = 0, \\ V_{i,j}^\top \phi_{i,j}(\varphi_{i,j-1}(v)), & j \in \{1, \ldots, k_i\}, \end{cases} \tag{1}$$

with $\Phi_i(v, \theta_i) = \varphi_{i,k_i}(v)$.

For each $j \in \{0, 1, \ldots, k_i\}$, the matrix $V_{i,j} \in \mathbb{R}^{L_{i,j}^a \times L_{i,j+1}}$ contains the weights and biases; in particular, if a layer has n inputs (including the appended bias) and the subsequent layer has m nodes, then $V \in \mathbb{R}^{n \times m}$ is constructed so that its $(r, c)^{\text{th}}$ entry represents the weight from the r^{th} node of the input to the c^{th} node of the output, with the last row corresponding to the bias terms. For the DNN architecture described by (1), the vector of DNN weights of the i^{th} block is $\theta_i \triangleq \begin{bmatrix} \operatorname{vec}(V_{i,0})^\top & \cdots & \operatorname{vec}(V_{i,k_i})^\top \end{bmatrix}^\top \in \mathbb{R}^{p_i}$, where $p_i \triangleq \sum_{j=0}^{k_i} L_{i,j}^a L_{i,j+1}$. For all $(i, j) \in \{0, \ldots, b\} \times \{0, \ldots, k_i\}$, the ensemble activation function of the j^{th} layer of the i^{th} block is given by $\phi_{i,j} : \mathbb{R}^{L_{i,j}} \to \mathbb{R}^{L_{i,j}^a}$, where $\phi_{i,j}(\varphi_{i,j-1}) \triangleq \begin{bmatrix} \varsigma_{i,j}((\varphi_{i,j-1})_1) & \varsigma_{i,j}((\varphi_{i,j-1})_2) & \cdots & \varsigma_{i,j}\left((\varphi_{i,j-1})_{L_{i,j}}\right) & 1 \end{bmatrix}^\top \in \mathbb{R}^{L_{i,j}^a}$, the activation function is $\varsigma_{i,j} : \mathbb{R} \to \mathbb{R}$, and 1 accounts for the bias term.

A pre-activation design is used so that, before each block (except block 0), the output of the previous block is processed by an external activation function. Specifically, for each block $i \in \{1, \ldots, b\}$, define the pre-activation mapping $\psi_i : \mathbb{R}^{L_{i,k_i+1}} \to \mathbb{R}^{L_{i,k_i+1}^a}$ by $\psi_i(\kappa_i) = [\varrho_i((\kappa_i)_1) \;\; \varrho_i((\kappa_i)_2) \;\; \cdots$

$\varrho_i\left((\kappa_i)_{L_{i,k_i+1}}\right) \quad 1]^\top \in \mathbb{R}^{L^a_{i,k_i+1}}$ where $(\kappa_i)_\ell$ denotes the ℓ^{th} component of κ_i, the activation function is $\varrho_i : \mathbb{R} \to \mathbb{R}$, and 1 accounts for the bias term. The output of ψ_i serves as the input to block i and the residual connection is implemented by adding the current block output to the pre-activated output from the previous block. Hence, the ResNet recursion is defined by

$$\kappa_{i+1} \triangleq \begin{cases} \Phi_0\left(\kappa_0^a, \theta_0\right), & i = 0, \\ \kappa_i + \Phi_i\left(\psi_i\left(\kappa_i\right), \theta_i\right), & i \in \{1, \ldots, b\}, \end{cases} \tag{2}$$

with output $y \in \mathbb{R}^{L_{\text{out}}}$ and overall parameter vector $\Theta \triangleq \begin{bmatrix} \theta_0^\top & \cdots & \theta_b^\top \end{bmatrix}^\top \in \mathbb{R}^{\mathtt{p}}$, with $\mathtt{p} \triangleq \sum_{i=0}^{b} p_i$, where $\kappa_0^a \triangleq \begin{bmatrix} \kappa_0^\top & 1 \end{bmatrix}^\top \in \mathbb{R}^{L^a_{0,0}}$ denotes the augmented input to block $i = 0$. Therefore, the complete ResNet is represented as $\Psi : \mathbb{R}^{L_{\text{in}}} \times \mathbb{R}^{\mathtt{p}} \to \mathbb{R}^{L_{\text{out}}}$ expressed as $\Psi(x, \Theta) = \kappa_{b+1}$.

In order for ResNets to act as universal approximators of continuous functions on compact domains, their activation functions must be selected to belong to a class of functions called $\mathcal{A}_{\text{quad}}$.

Definition 1. *[3, Definition 3.1] The activation function $\varsigma : \mathbb{R} \to \mathbb{R}$ is said to be of class $\mathcal{A}_{quad}$ (denoted $\varsigma \in \mathcal{A}_{quad}$) if it is (i) nondecreasing, (ii) non-constant, and (iii) there exists an $\ell \in \mathbb{Z}_{\geq 0}$ such that ς is a globally-defined solution to $\frac{d\zeta}{ds} = a_0 + a_1\zeta + a_2\zeta^2$, where $\zeta \triangleq \frac{d^{(\ell)}\varsigma}{ds^{(\ell)}}$ is injective, $a_0, a_1, a_2 \in \mathbb{R}$, $a_2 \neq 0$, and $\frac{d^{(0)}\varsigma}{ds^{(0)}} = \varsigma$.*[1]

Theorem 1. *[3, Corollary 5.2] (Universal Approximation Theorem for Residual Neural Networks). For any compact domain $\Omega \subsetneq \mathbb{R}^{L_{in}}$, continuous function $F : \Omega \to \mathbb{R}^{L_{out}}$, and approximation accuracy $\overline{\varepsilon} \in \mathbb{R}_{>0}$, there exists a number of blocks $b \in \mathbb{Z}_{\geq 1}$ (thus, a $\mathtt{p} \in \mathbb{Z}_{\geq 1}$) and a parameter $\theta^* \in \mathbb{R}^{\mathtt{p}}$ such that for a ResNet architecture $\Psi : \mathbb{R}^{L_{in}} \times \mathbb{R}^{\mathtt{p}} \to \mathbb{R}^{L_{out}}$ with activation functions $\varsigma_{i,j} \in \mathcal{A}_{quad}$ for all $(i,j) \in \{0, \ldots, b\} \times \{0, \ldots, k_i\}$, blocks of width $w = \max\left(L_{in}, L_{out}\right) + 1$ (i.e., $L_{i,j} \geq w$ for all $(i,j) \in \{0, \ldots, b\} \times \{1, \ldots, k_i\}$) and depth 1 (i.e., $k_i \geq 1$ for all $i \in \{0, \ldots, b\}$),*

$$\sup_{x \in \Omega} \left\| \varepsilon\left(x, \theta^*\right) \right\| \leq \overline{\varepsilon}, \tag{3}$$

where $\varepsilon\left(x, \theta^*\right) \triangleq F(x) - \Psi\left(x, \theta^*\right)$.

HAVE TO MENTION HIGHER-ORDER APPROXIMATION.

III. Problem Formulation

Consider the second-order nonlinear dynamical system described by the differential equation,

$$\ddot{q}(t) = f\left(q(t), \dot{q}(t)\right) + u(t) + d(t), \tag{4}$$

where $t \in \mathbb{R}_{\geq 0}$ denotes time, $u : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ denotes the control input, $f : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ denotes unknown drift dynamics of class $\mathcal{C}^2$, and $d : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ denotes an unknown, time-variant exogenous disturbance of class $\mathcal{C}^2$. For an initial condition set $(t_0, q_0, \dot{q}_0) \in \mathbb{R}_{\geq 0} \times \mathbb{R}^n \times \mathbb{R}^n$, let $q : \mathcal{I} \to \mathbb{R}^n$, $\dot{q} : \mathcal{I} \to \mathbb{R}^n$, and $\ddot{q} : \mathcal{I} \to \mathbb{R}^n$ denote the generalized position, velocity, and acceleration, respectively, of the resulting solution to (4) such that $q(t_0) = q_0$, $\dot{q}(t_0) = \dot{q}_0$, and $\mathcal{I} \triangleq [0, T_{\max})$ is the interval of existence for $T_{\max} \in (t_0, \infty]$.

The desired trajectory for (4) is governed by the autonomous second-order system,

$$\ddot{q}_{\mathrm{d}}(t) = h\left(q_{\mathrm{d}}(t), \dot{q}_{\mathrm{d}}(t)\right), \tag{5}$$

where $h : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ is an unknown function of class $\mathcal{C}^2$. For an initial condition set $(t_0, q_{\mathrm{d},0}, \dot{q}_{\mathrm{d},0}) \in \mathbb{R}_{\geq 0} \times \mathbb{R}^n \times \mathbb{R}^n$, let $q_{\mathrm{d}} : \mathbb{R}_{\geq t_0} \to \mathbb{R}^n$, $\dot{q}_{\mathrm{d}} : \mathbb{R}_{\geq t_0} \to \mathbb{R}^n$ and $\ddot{q}_{\mathrm{d}} : \mathbb{R}_{\geq t_0} \to \mathbb{R}^n$ denote generalized desired position, velocity, and acceleration, respectively, of the resulting solution to (5) such that $q_{\mathrm{d}}(t_0) = q_{\mathrm{d},0}$ and $\dot{q}_{\mathrm{d}}(t_0) = \dot{q}_{\mathrm{d},0}$.

Let the tracking error be $e(t) \triangleq q_{\mathrm{d}}(t) - q(t)$. The control objective is to design a controller such that $\|e\|$ converges exponentially to zero despite the presence of non-vanishing disturbances and unknown dynamics. Since $\ddot{q}_{\mathrm{d}}$ is unknown and h is time-invariant, the objective is said to be target tracking rather than trajectory tracking. The subsequent assumptions ensure feasibility of this objective.

Assumption 1. [4, Assumption 1]. There exist known constants $\overline{q}_{\mathrm{d}}, \overline{\dot{q}}_{\mathrm{d}}, \overline{\ddot{q}}_{\mathrm{d}}, \overline{\dddot{q}}_{\mathrm{d}} \in \mathbb{R}_{\geq 0}$ such that $\|q_{\mathrm{d}}(t)\| \leq \overline{q}_{\mathrm{d}}$, $\|\dot{q}_{\mathrm{d}}(t)\| \leq \overline{\dot{q}}_{\mathrm{d}}$, $\|\ddot{q}_{\mathrm{d}}(t)\| \leq \overline{\ddot{q}}_{\mathrm{d}}$, and $\|\dddot{q}_{\mathrm{d}}(t)\| \leq \overline{\dddot{q}}_{\mathrm{d}}$ for all $t \in \mathbb{R}_{\geq 0}$.

Assumption 2. [4]. There exist known constants $\overline{d}, \overline{\dot{d}}, \overline{\ddot{d}} \in \mathbb{R}_{\geq 0}$ such that $\|d(t)\| \leq \overline{d}$, $\left\|\dot{d}(t)\right\| \leq \overline{\dot{d}}$, and $\left\|\ddot{d}(t)\right\| \leq \overline{\ddot{d}}$ for all $t \in \mathbb{R}_{\geq 0}$.[2]

IV. Control Design

To facilitate subsequent control design, the first and second auxiliary tracking errors are defined as

$$r_1(t) \triangleq \dot{e}(t) + k_1 e(t), \tag{6}$$

$$r_2(t) \triangleq \dot{r}_1(t) + k_2 r_1(t), \tag{7}$$

where $k_1, k_2 \in \mathbb{R}_{>0}$ are user-selected gains. Differentiating (6) and using (4)-(6) gives

$$\begin{aligned} \dot{r}_1(t) = {} & h\left(q_{\mathrm{d}}(t), \dot{q}_{\mathrm{d}}(t)\right) - f\left(q(t), \dot{q}(t)\right) \\ & - u(t) - d(t) + k_1 r_1(t) - k_1^2 e(t), \end{aligned} \tag{8}$$

for all $t \in \mathcal{I}$. Differentiating (7) and substituting (8) and (6) gives

$$\begin{aligned} \dot{r}_2(t) = {} & F(\kappa(t)) - \left.\frac{\partial f}{\partial \dot{q}}\right|_{(q(t), \dot{q}(t))} d(t) - \dot{u}(t) - \dot{d}(t) \\ & + (k_1 + k_2) r_2(t) - \left(k_1 k_2 + k_2^2 + k_1^2\right) r_1(t) + k_1^3 e(t), \end{aligned} \tag{9}$$

for all $t \in \mathcal{I}$, where $\kappa : \mathcal{I} \to \mathbb{R}^{5n}$ is defined as $\kappa(t) \triangleq \begin{bmatrix} q^\top(t) & \dot{q}^\top(t) & q_{\mathrm{d}}(t) & \dot{q}_{\mathrm{d}}(t) & u^\top(t) \end{bmatrix}^\top$ and $F : \mathbb{R}^{5n} \to \mathbb{R}^n$ is defined as $F(\kappa) \triangleq \left.\frac{\partial h}{\partial q_{\mathrm{d}}}\right|_{(q_{\mathrm{d}}, \dot{q}_{\mathrm{d}})} \dot{q}_{\mathrm{d}} + \left.\frac{\partial h}{\partial \dot{q}_{\mathrm{d}}}\right|_{(q_{\mathrm{d}}, \dot{q}_{\mathrm{d}})} h\left(q_{\mathrm{d}}, \dot{q}_{\mathrm{d}}\right) - \left.\frac{\partial f}{\partial q}\right|_{(q, \dot{q})} \dot{q} - \left.\frac{\partial f}{\partial \dot{q}}\right|_{(q, \dot{q})} f(q, \dot{q}) - \left.\frac{\partial f}{\partial \dot{q}}\right|_{(q, \dot{q})} u$.

[1] By consequence of Definition 1, any $\varsigma \in \mathcal{A}_{\text{quad}}$ is necessarily an analytical function. Thus, ς and all its derivatives are locally Lipschitz on $\mathbb{R}$. Hyperbolic tangent, sigmoid, and softplus are of class $\mathcal{A}_{\text{quad}}$ but ReLU, swish, and the identity mapping are not. The function $\varsigma(s) = \frac{1}{c}\log\left(1 + \mathrm{e}^{cs}\right)$ approximates ReLU for large $c \in \mathbb{R}_{>0}$ and is of class $\mathcal{A}_{\text{quad}}$. Leaky ReLU is given as $\varsigma(s) = s$ for all $s \in \mathbb{R}_{\geq 0}$ and $\varsigma(s) = rs$ (with $r \in \mathbb{R}_{>0}$) for all $s < 0$. The function $\varsigma(s) = rs + \frac{1}{c}\log\left(1 + \mathrm{e}^{(1-r)cs}\right)$ approximates leaky ReLU for large $c \in \mathbb{R}_{>0}$ and is of class $\mathcal{A}_{\text{quad}}$.

[2] While adaptive RISE techniques like [5] could be used to compensate for unknown bounds $\overline{d}, \overline{\dot{d}}, \overline{\ddot{d}}$, the described formulation therein yields asymptotic convergence of the tracking error. Achieving exponential convergence without Assumption 2 remains an open challenge.

A. Function Approximation

Since f and h are unknown, F is unknown and will be approximated online with a user-selected parametric universal approximator as $\Phi\left(\kappa,\widehat{\theta}\right)$ where $\Phi:\mathbb{R}^{L_{\text{in}}}\times\mathbb{R}^{\mathtt{p}}\to\mathbb{R}^{L_{\text{out}}}$ is a function approximator architecture mapping of class $\mathcal{C}^1$, $\widehat{\theta}\in\mathbb{R}^{\mathtt{p}}$ is a vectorized set of parameters, $L_{\text{out}}=n$, $L_{\text{in}}=5n$, and $\mathtt{p}\in\mathbb{Z}_{\geq 1}$ is the number of parameters. In this work, any class $\mathcal{C}^1$ parametric approximator which satisfies a universal approximation theorem like Theorem 1 can be used for Φ, like radial basis functions, ResNets, transformers (e.g., [6]), or LSTMs (e.g., [7]). The parameter $\widehat{\theta}$ is intended to dynamically estimate θ^* and evolve continuously but be constrained to a user-defined compact search space $\mho\subseteq\overline{B}_{\overline{\theta}}\left(0_{\mathtt{p}}\right)$, where $\overline{\theta}\in\mathbb{R}_{>0}$. The rule governing the trajectory of $t\mapsto\widehat{\theta}(t)$ is called the update law and is chosen by the user but must satisfy the following properties.

Property 1. [8, Assumption 1] The parameter's trajectory $\widehat{\theta}:\mathbb{R}_{\geq 0}\to\mathbb{R}^{\mathtt{p}}$ is of class $\mathcal{C}^1$ and there exist known constants $\overline{\theta},\overline{\dot{\theta}}\in\mathbb{R}_{>0}$ such that the trajectory and its first time derivative $\dot{\widehat{\theta}}:\mathbb{R}_{\geq 0}\to\mathbb{R}^{\mathtt{p}}$ satisfy $\left\|\widehat{\theta}\left(t\right)\right\|\leq\overline{\theta}$ and $\left\|\dot{\widehat{\theta}}\left(t\right)\right\|\leq\overline{\dot{\theta}}$ for all $t\in\mathbb{R}_{\geq 0}$.

Property 2. There exists a known constant $\overline{\varepsilon}_{\text{d}}\in\mathbb{R}_{>0}$ such that parameter's trajectory satisfies $E_{\Omega}\left(\widehat{\theta}\left(t\right)\right)\leq\overline{\varepsilon}_{\text{d}}$ for all $t\in\mathbb{R}_{\geq 0}$, where the reconstruction error is defined as

$$\varepsilon\left(\kappa,\theta\right)\triangleq\Phi\left(\kappa,\theta\right)-F\left(\kappa\right), \tag{10}$$

the approximation accuracy is $E_{\Omega}\left(\theta\right)\triangleq\sup_{\kappa\in\Omega}\left\|\varepsilon\left(\kappa,\theta\right)\right\|+\sup_{\kappa\in\Omega}\left\|\frac{\partial\varepsilon}{\partial\kappa}\big|_{(\kappa,\theta)}\right\|$, and the desired domain where the approximation of $F\left(\kappa\right)$ by $\Phi\left(\kappa,\widehat{\theta}\right)$ must hold is

$$\Omega\triangleq\left\{\zeta\in\mathbb{R}^{5n}:\left\|\zeta\right\|\leq R_{\Omega}\right\}, \tag{11}$$

for $R_{\Omega}\in\mathbb{R}_{>0}$.

Remark 1. Since the design of the update law and the choice of parametric approximator for Φ are at the user's discretion, the approach in this work is said to be modular.

Property 1 can be met, for instance, by using a projection algorithm and limiting the speed of parameter changes. Satisfying Property 2 requires careful consideration of both parameter initialization and the update law. The initial parameter $\widehat{\theta}\left(t_0\right)$ must either be obtained by pre-training the function approximator on input-output pairs generated by a function representative of F or by using the following approach. Notice that extreme value theorem (EVT) ensures that $\left\|F\right\|+\left\|\frac{dF}{d\kappa}\right\|$ achieves a finite maximum over Ω since Ω is compact and F is of class $\mathcal{C}^1$. Assume a bound $\overline{F}\in\mathbb{R}_{\geq 0}$ on such a maximum is known. Pick $\widehat{\theta}\left(t_0\right)$ and a suitable $\mho$ such that $\Phi\left(\kappa,\widehat{\theta}\left(t_0\right)\right)=0_n$ for all $\kappa\in\Omega$. Finally, select $\overline{\varepsilon}_{\text{d}}=\overline{F}$. Satisfying Property 2 beyond the initial time requires that any information gained online to refine the parameter not result in a $E_{\Omega}\left(\widehat{\theta}\left(t\right)\right)$ that exceeds $\overline{\varepsilon}_{\text{d}}$. Heuristically, this phenomena, called catastrophic forgetting, can be avoided by using experience replay methods like concurrent learning (CL).[3]

[3]For example, a first-order CL-based update law for parameters in universal approximators like Φ can be found in [9]. Note that any discrete updates to a history stack made online may not satisfy the $\mathcal{C}^1$ requirement of Property 1. However, online history stack management is not required to prevent catastrophic forgetting.

The expressive powers of ResNets and many similar approximators rely on the ability to select θ^* from the whole of $\mathbb{R}^{\mathtt{p}}$ (cf. Theorem 1) to approximate any continuous function. Since in practice $\widehat{\theta}$ must be constrained to lie in a prescribed compact search space $\mho$, it is assumed a suitable θ^* satisfying Property 2 also lies in $\mho$. The following assumption formalizes this requirement and the need for simultaneous approximation of $\frac{dF}{d\kappa}$ in subsequent analysis.

Assumption 3. The function F and its first derivative $\frac{dF}{d\kappa}$ can be uniformly approximated to arbitrary precision using bounded weights for a sufficiently large approximator architecture. Formally, $F\in\{\mathcal{C}^1(\Omega;\mathbb{R}^n):\forall\overline{\varepsilon}\in\mathbb{R}_{>0},\exists\mathtt{p}\in\mathbb{Z}_{\geq 1}:\exists\mho\subsetneq\mathbb{R}^{\mathtt{p}}:\exists\theta^*\in\mho:E_{\Omega}(\theta^*)\leq\overline{\varepsilon}\}\subsetneq\mathcal{C}^1(\Omega;\mathbb{R}^n)$.

Assumption 3 is required because in parametric nonlinear regression, the user can only pick two of the following three desirable qualities for their approximator: (i) F can be any sufficiently smooth (e.g., of class $\mathcal{C}^1$) function, (ii) F can be approximated arbitrarily well ($\overline{\varepsilon}_{\text{d}}\to 0$ as $\mathtt{p}\to\infty$), and (iii) the weights needed to achieve the desired accuracy for the target function lies in a prescribed set $\mho$ with a known circumradius $\overline{\theta}$. Without (ii), there will exist sufficiently small values of $\overline{\varepsilon}_{\text{d}}$ that are not achievable for the given $\mho$, F, and Ω even as $\mathtt{p}\to\infty$. That is, the infimum of all values $\overline{\varepsilon}_{\text{d}}$ may achieve is an unknown positive value (rather than zero) that might be arbitrarily large. Thus, (ii) is required. Since computers possess finite memory to store $\widehat{\theta}$, (iii) is required. Subsequent analysis also relies on (iii). Thus, (i) must be sacrificed in place of Assumption 3. Under Assumption 3, the set $\Theta^*\triangleq\arg\min_{\theta\in\mho}E_{\Omega}\left(\theta\right)$ is nonempty and all $\theta^*\in\Theta^*$ satisfy $\left\|\theta^*\right\|\leq\overline{\theta}$.

B. Control Law

Motivated by the form of (9), the controller is designed as

$$\begin{aligned}\dot{u}\left(t\right)=&\left(k_1+k_2+k_3\right)r_2\left(t\right)+k_1^3e\left(t\right)\\&+\left(1-k_1^2-k_2^2-k_1k_2\right)r_1\left(t\right)\\&+K_{\text{RISE}}\text{sgn}\left(r_1\left(t\right)\right)+\Phi\left(\kappa\left(t\right),\hat{\theta}\left(t\right)\right),\end{aligned} \tag{12}$$

where $K_{\text{RISE}},k_3\in\mathbb{R}_{>0}$ are user-defined constants. Integrating both sides of (12) and picking the initial condition $u\left(t_0\right)=K_{\text{P}}e\left(t_0\right)+K_{\text{D}}\dot{e}\left(t_0\right)$ yields the implementable form[4]

$$u\left(t\right)=u_{\text{PID}}\left(t\right)+u_{\text{RISE}}\left(t\right)+\int_{t_0}^{t}\left(\Phi\left(\kappa\left(\tau\right),\hat{\theta}\left(\tau\right)\right)\right)d\tau, \tag{13}$$

where

$$u_{\text{PID}}\left(t\right)\triangleq K_{\text{P}}e\left(t\right)+K_{\text{D}}\dot{e}\left(t\right)+K_{\text{I}}\int_{t_0}^{t}e\left(\tau\right)d\tau,$$

$$u_{\text{RISE}}\left(t\right)\triangleq K_{\text{RISE}}\int_{t_0}^{t}\text{sgn}\left(r_1\left(\tau\right)\right)d\tau,$$

$$\begin{aligned}K_{\text{P}}&\triangleq k_1k_2+k_1k_3+k_2k_3+1,\\K_{\text{I}}&\triangleq k_1k_2k_3+k_1,\\K_{\text{D}}&\triangleq k_1+k_2+k_3.\end{aligned}$$

[4]Although $\ddot{q}$ is not assumed to be measurable, it appears in (12) by virtue of r_2. However, note that the implementable form in (13) only requires q and $\dot{q}$.

Define the idealized input $\kappa_{\rm d} : \mathbb{R}_{\geq 0} \to \mathbb{R}^{5n}$ seen when (4) perfectly tracks (5) as $\kappa_{\rm d} \triangleq \begin{bmatrix} q_{\rm d}^\top & \dot{q}_{\rm d}^\top & q_{\rm d}^\top & \dot{q}_{\rm d}^\top & u_{\rm d}^\top \end{bmatrix}^\top \in \mathbb{R}^{5n}$, where $u_{\rm d} : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ is the ideal tracking component of the controller obtained from feedback linearization,

$$u_{\rm d}(t) \triangleq d(t) + f(q_{\rm d}(t), \dot{q}_{\rm d}(t)) - \ddot{q}_{\rm d}(t). \tag{14}$$

Substituting (12) and (10) into (9), using the definition of $\kappa_{\rm d}$ and performing algebraic manipulation gives that $r_2 : \mathcal{I} \to \mathbb{R}^n$ is a Filippov solution to

$$\begin{aligned} \dot{r}_2(t) \in \{ &\widetilde{N}\left(\kappa, \kappa_{\rm d}, \theta^*, \hat{\theta}, t\right) - k_3 r_2(t) - r_1(t) - K_{\rm RISE}\nu_1 \\ &+ D\left(\kappa_{\rm d}, \theta^*, \hat{\theta}, t\right) : \nu_1 \in K[\operatorname{sgn}](r_1)\}, \end{aligned} \tag{15}$$

where

$$\begin{aligned} D_t &\triangleq -\dot{d}(t) + \varepsilon(\kappa_{\rm d}(t), \theta^*) \\ &\quad + N_0\left(\kappa_{\rm d}(t), \theta^*, \hat{\theta}(t)\right) \\ &\quad - \left.\frac{\partial f}{\partial \dot{q}}\right|_{(q_{\rm d}(t), \dot{q}_{\rm d}(t))} d(t), \\ N_0\left(s, \theta^*, \hat{\theta}\right) &\triangleq \Phi(s, \theta^*) - \Phi\left(s, \hat{\theta}\right), \\ \widetilde{N}_t &\triangleq \widetilde{N}_1\left(\kappa(t), \kappa_{\rm d}(t), \theta^*, \hat{\theta}(t)\right) \\ &\quad + \widetilde{N}_\varepsilon(\kappa(t), \kappa_{\rm d}(t)) \\ &\quad + \widetilde{N}_2(q(t), \dot{q}(t), q_{\rm d}(t), \dot{q}_{\rm d}(t))\, d(t), \\ \widetilde{N}_\varepsilon(\kappa, \kappa_{\rm d}, \theta^*) &\triangleq \varepsilon(\kappa, \theta^*) - \varepsilon(\kappa_{\rm d}, \theta^*), \\ \widetilde{N}_1\left(\kappa, \kappa_{\rm d}, \theta^*, \hat{\theta}\right) &\triangleq N_0\left(\kappa, \theta^*, \hat{\theta}\right) - N_0\left(\kappa_{\rm d}, \theta^*, \hat{\theta}\right), \\ \widetilde{N}_2(q, \dot{q}, q_{\rm d}, \dot{q}_{\rm d}) &\triangleq \left.\frac{\partial f}{\partial \dot{q}}\right|_{(q_{\rm d}, \dot{q}_{\rm d})} - \left.\frac{\partial f}{\partial \dot{q}}\right|_{(q, \dot{q})}, \end{aligned}$$

Since f is $\mathcal{C}^1$, Assumption 1 and EVT guarantee that there exist constants $\overline{f}, \overline{\dot{f}} \in \mathbb{R}_{\geq 0}$ such that $\|f(q_{\rm d}(t), \dot{q}_{\rm d}(t))\| \leq \overline{f}$ and $\left\|\dot{f}(q_{\rm d}(t), \dot{q}_{\rm d}(t))\right\| \leq \overline{\dot{f}}$ for all $t \in \mathbb{R}_{\geq 0}$.

Assumption 4. The constants $\overline{f}, \overline{\dot{f}} \in \mathbb{R}_{\geq 0}$ are known.

By Assumption 4, there exist known bounds $\overline{u}_{\rm d} \triangleq \overline{d} + \overline{f} + \overline{\ddot{q}}_{\rm d}$ and $\overline{\dot{u}}_{\rm d} \triangleq \overline{\dddot{q}}_{\rm d} + \overline{\dot{f}}\overline{\dot{q}}_{\rm d} + \overline{\dot{f}}\overline{\ddot{q}}_{\rm d} + \overline{\dot{d}}$ such that $\|u_{\rm d}(t)\| \leq \overline{u}_{\rm d}$ and $\|\dot{u}_d(t)\| \leq \overline{\dot{u}}_{\rm d}$ for all $t \in \mathbb{R}_{\geq 0}$. By Assumptions 1 and 4, there exist known constants $\overline{\kappa}_{\rm d} \triangleq 2\overline{q}_{\rm d} + 2\overline{\dot{q}}_{\rm d} + \overline{u}_{\rm d}$ and $\overline{\dot{\kappa}}_{\rm d} \triangleq 2\overline{\dot{q}}_{\rm d} + 2\overline{\ddot{q}}_{\rm d} + \overline{\dot{u}}_{\rm d}$ such that $\|\kappa_{\rm d}(t)\| \leq \overline{\kappa}_{\rm d}$ and $\|\dot{\kappa}_{\rm d}(t)\| \leq \overline{\dot{\kappa}}_{\rm d}$ for all $t \in \mathbb{R}_{\geq 0}$. Let the ensemble error state $z : \mathcal{I} \to \mathbb{R}^{3n}$ be defined as $z(t) \triangleq \begin{bmatrix} e^\top(t) & r_1^\top(t) & r_2^\top(t) \end{bmatrix}^\top$. Since f is $\mathcal{C}^2$, EVT, Assumption 1, and [10, Lemma 5] guarantee that there exist functions $\rho_f, \rho_{f'} : \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ that are class $\mathcal{P}_\infty$ with respect to each argument such that $\|f(q, \dot{q}) - f(q_{\rm d}, \dot{q}_{\rm d})\| \leq \rho_f(k_1, \|z\|)\|z\|$ and $\left\| \left.\frac{\partial f}{\partial \dot{q}}\right|_{(q_{\rm d}, \dot{q}_{\rm d})} - \left.\frac{\partial f}{\partial \dot{q}}\right|_{(q, \dot{q})} \right\| \leq \rho_{f'}(k_1, \|z\|)\|z\|$.

Assumption 5. The functions ρ_f and $\rho_{f'}$ are known.

Define $\widetilde{u}(t) \triangleq u_{\rm d}(t) - u(t)$. Subtracting (14) from (4) and using (6), (7), and the time derivative of (7) gives that

$$\begin{aligned} \widetilde{u}(t) = &\, r_2(t) - k_1\dot{e}(t) - k_2 r_1(t) \\ &- f(q_{\rm d}(t), \dot{q}_{\rm d}(t)) + f(q(t), \dot{q}(t)), \end{aligned} \tag{16}$$

for all $t \in \mathcal{I}$. By Assumption 5, there exists a known function $\rho_{\widetilde{u}} : \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ that is of class $\mathcal{P}_\infty$ with respect to each argument such that $\|\widetilde{u}\| \leq \rho_{\widetilde{u}}(k_1, k_2, \|z\|)\|z\|$ where $\rho_{\widetilde{u}}(k_1, k_2, s) \triangleq 1 + k_1 + k_1^2 + k_2 + \rho_f(k_1, s)$. By (6), Assumption 5, the definition of $\rho_{\widetilde{u}}$, and the fact that $\|\kappa_{\rm d}\|$ is bounded by a known constant, there exists a known function that is of class $\mathcal{P}_\infty$ with respect to each argument such that $\|\widetilde{\kappa}\| \leq \rho_{\widetilde{\kappa}}(k_1, k_2, \|z\|)\|z\|$, where $\widetilde{\kappa}(t) \triangleq \kappa_{\rm d}(t) - \kappa(t)$ and $\rho_{\widetilde{\kappa}}(k_1, k_2, s) \triangleq k_1 + 2 + \rho_{\widetilde{u}}(k_1, k_2, s)$. Since Φ is of class $\mathcal{C}^1$ and $\|\kappa_{\rm d}\|$ is bounded, using [10, Lemma 5], the bound on $\|\widetilde{\kappa}\|$ and Property 3, there exists a known function $\rho_\Phi : \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ that is of class $\mathcal{P}_\infty$ with respect to each argument such that $\left\|\widetilde{N}_1\left(\kappa, \kappa_{\rm d}, \theta^*, \hat{\theta}\right)\right\| \leq \rho_\Phi(k_1, k_2, \|z\|)\|z\|$. Choose $R_\Omega = \overline{\kappa}$, where $\overline{\kappa} \triangleq \overline{\kappa}_{\rm d} + \rho_{\widetilde{\kappa}}(k_1, k_2, C_\Delta) C_\Delta$, where $C_\Delta \in \mathbb{R}_{>0}$ is subsequently defined to ensure that any $t \mapsto z(t)$ satisfies $z(t) \in \mathcal{D} \triangleq \left\{\zeta \in \mathbb{R}^{3n} : \|\zeta\| \leq C_\Delta\right\}$. Recall $\sup(\mathcal{I}) = T_{\max}$. Let $T_{\rm exit} \in (t_0, \infty]$ denote the first time that the trajectory $t \mapsto z(t)$ escapes the set $\mathcal{D}$. Introduce $\mathcal{T} \triangleq \mathcal{I} \cap [t_0, T_{\rm exit})$. By the choice of R_Ω, $\overline{B}_{\overline{\kappa}_{\rm d}}(0_{5n}) \subseteq \Omega$ and $\overline{\kappa}_{\rm d} \leq \overline{\kappa}$. Using this fact, Property 2 ensures that $\|\varepsilon(\kappa_{\rm d}(t))\| \leq \overline{\varepsilon}_{\rm d}$ and $\left\| \left.\frac{d\varepsilon}{d\kappa}\right|_{\kappa_{\rm d}(t)} \right\| \leq \overline{\varepsilon}_{\rm d}$ for all $t \in \mathcal{T}$. Since $\|\kappa\| \leq \overline{\kappa}_{\rm d} + \|\widetilde{\kappa}\|$ by triangle inequality, the bound on $\|\widetilde{\kappa}\|$ gives that $\|\kappa(t)\| \leq \overline{\kappa}$ for all $t \in \mathcal{T}$. Since ε is of class $\mathcal{C}^1$, it follows that $\left\|\widetilde{N}_\varepsilon(\kappa(t), \kappa_{\rm d}(t))\right\| \leq \overline{\varepsilon}_{\rm d}\rho_{\widetilde{\kappa}}(k_1, k_2, \|z(t)\|)\|z(t)\|$ for all $t \in \mathcal{T}$ [10, Lemma 5]. Since f is of class $\mathcal{C}^2$, Assumption 1 and EVT guarantee that there exist constants $\overline{f'}, \overline{f''} \in \mathbb{R}_{\geq 0}$ such $\left\| \left.\frac{\partial f}{\partial \dot{q}}\right|_{(q_{\rm d}(t), \dot{q}_{\rm d}(t))} \right\| \leq \overline{f'}$, $\left\| \left.\frac{\partial f}{\partial \dot{q}\partial q}\right|_{(q_{\rm d}(t), \dot{q}_{\rm d}(t))} \right\| \leq \overline{f''}$, and $\left\| \left.\frac{\partial^2 f}{\partial \dot{q}^2}\right|_{(q_{\rm d}(t), \dot{q}_{\rm d}(t))} \right\| \leq \overline{f''}$ for all $t \in \mathbb{R}_{\geq 0}$.

Assumption 6. The constants $\overline{f'}$ and $\overline{f''}$ are known.

By Assumption 5, $\left\|\widetilde{N}_2(q, \dot{q}, q_{\rm d}, \dot{q}_{\rm d})\right\| \leq \rho_{f'}(k_1, \|z\|)\|z\|$. Using Assumption 2, it follows that there exists a known function $\rho_0 : \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ that is of class $\mathcal{P}_\infty$ with respect to each argument such that $\left\|\widetilde{N}_t\right\| \leq \rho_0(k_1, k_2, \|z(t)\|)\|z(t)\|$ for all $t \in \mathcal{T}$, where $\rho_0(k_1, k_2, s) \triangleq \rho_\Phi(k_1, k_2, s) + \overline{\varepsilon}_{\rm d}\rho_{\widetilde{\kappa}}(k_1, k_2, s) + \overline{d}\rho_{f'}(k_1, s)$.

V. Stability Analysis

From (6), (7), and (15), the evolution of z is a Filippov solution to the initial value problem,

$$\dot{z} \in G(t, z), \quad z(t_0) = z_0, \tag{17}$$

where $z_0 \in \mathbb{R}^{3n}$ is the initial ensemble error and $G : \mathbb{R}^{3n} \times \mathbb{R}_{\geq 0} \rightrightarrows \mathbb{R}^{3n}$ is the set-valued map

$$\begin{aligned} G(z, t) \triangleq \Bigg\{ &\begin{bmatrix} r_1 - k_1 e \\ r_2 - k_2 r_1 \\ \widetilde{N}\left(\kappa, \kappa_{\rm d}, \theta^*, \hat{\theta}, t\right) - k_3 r_2 - r_1 - K_{\rm RISE}\nu + D_t \end{bmatrix} \\ &: \nu \in K[\operatorname{sgn}](r_1) \Bigg\}. \end{aligned}$$

To establish exponential convergence to the origin for the system given by (17), an auxiliary function called a P-function is introduced. Sufficient conditions to ensure that the P-function is nonnegative are presented next.

Consider an arbitrary Filippov solution $t \mapsto z(t)$ of (17) on $\mathcal{T}$. Define $\zeta(t) \triangleq K_{\rm RISE}\|r_1(t)\|_1 - r_1^\top(t) D_t$ and $\phi(t) \triangleq r_1^\top(t)\dot{D}_t + (k_2 - \lambda_P)\zeta(t)$, where $\lambda_P \in \mathbb{R}_{>0}$. Since r_1 is absolutely continuous on $\mathcal{T}$ and $\|\cdot\|_1$ is globally Lipschitz, $t \mapsto \|r_1(t)\|_1$ is absolutely continuous; since $t \mapsto D_t$ is absolutely continuous, ζ is absolutely continuous and ϕ is measurable and locally integrable on $\mathcal{T}$. Let $\eta : \mathcal{T} \to \mathbb{R}$ be the unique absolutely continuous solution of $\dot{\eta}(t) = -\lambda_P \eta(t) + \phi(t)$ with $\eta(t_0) = 0$, and define the P-function $P : \mathcal{T} \to \mathbb{R}$ as

$$P(t) \triangleq \zeta(t) + \eta(t). \tag{18}$$

By the nonsmooth chain rule in [11, Theorem 2.2], $\frac{\mathrm{d}}{\mathrm{d}t}\|r_1(t)\|_1 \in \dot{r}_1^\top(t) K[\mathrm{sgn}](r_1(t))$ for almost all $t \in \mathcal{T}$, and $r_1^\top \nu = \|r_1\|_1$ for all $\nu \in K[\mathrm{sgn}](r_1)$. Therefore, differentiating (18) and using (7) and (18) yields

$$\dot{P} \in \left\{-\lambda_P P + r_2^\top(t)\left(K_{\mathrm{RISE}}\nu - D_t\right) : \nu \in K[\mathrm{sgn}](r_1(t))\right\}. \tag{19}$$

Proposition 1. *For the P-function defined in (18), if $K_{RISE} \geq \overline{D} + \frac{\overline{\dot{D}}}{k_2 - \lambda_P}$ where $k_2 > \lambda_P$, then $P(t) \geq 0$ for all $t \in \mathcal{T}$.*

Proof: Since the architecture mapping Φ, domain of approximation Ω, function F, and parameter search space $\mho$ are all time-invariant, Θ^* is also time-invariant. Thus, $\dot{\theta}^* = 0_{\mathrm{p}}$ for any $\theta^* \in \mho$. Using this fact, the continuity of Φ, and the fact that $\|\kappa_{\mathrm{d}}(t)\|$, $\|\dot{\kappa}_{\mathrm{d}}(t)\|$, $\|\varepsilon(\kappa_{\mathrm{d}}(t), \theta^*)\|$, and $\left\|\frac{\partial \varepsilon}{\partial \kappa}\big|_{(\kappa_{\mathrm{d}}(t),\theta^*)}\right\|$ are bounded by known constants for all $t \in \mathcal{T}$, EVT, Property 3, and Assumptions 1, 2, and 6 ensure that there exist known constants $\overline{D}, \overline{\dot{D}} \in \mathbb{R}_{\geq 0}$ such that $\|D_t\| \leq \overline{D}$ and $\left\|\dot{D}_t\right\| \leq \overline{\dot{D}}$ for all $t \in \mathcal{T}$. Using the Cauchy-Schwarz inequality and $\|r_1\| \leq \|r_1\|_1$ yields $-r_1^\top(t) D_t \geq -\|r_1(t)\|\|D_t\| \geq -\overline{D}\|r_1(t)\|_1$ and $r_1^\top(t)\dot{D}_t \geq -\|r_1(t)\|\left\|\dot{D}_t\right\| \geq -\overline{\dot{D}}\|r_1(t)\|_1$ for all $t \in \mathcal{T}$. Hence, $\zeta(t) \geq \left(K_{\mathrm{RISE}} - \overline{D}\right)\|r_1(t)\|_1$ and $\phi(t) \geq \left((k_2 - \lambda_P)\left(K_{\mathrm{RISE}} - \overline{D}\right) - \overline{\dot{D}}\right)\|r_1(t)\|_1$ for all $t \in \mathcal{T}$. If $K_{\mathrm{RISE}} \geq \overline{D} + \frac{\overline{\dot{D}}}{k_2 - \lambda_P}$ where $k_2 > \lambda_P$, then $(k_2 - \lambda_P)\left(K_{\mathrm{RISE}} - \overline{D}\right) - \overline{\dot{D}} > 0$, so $\phi(t) \geq 0$ for all $t \in \mathcal{T}$. The definition of $\dot{\eta}$ yields $\eta(t) = \int_{t_0}^t \mathrm{e}^{-\lambda_P(t-\tau)}\phi(\tau)\,\mathrm{d}\tau$. Because $\mathrm{e}^{-\lambda_P(t-\tau)} \geq 0$ for all $\tau \in [t_0, t]$ and $\phi(\tau) \geq 0$ for all $\tau \in [t_0, t]$, it follows that $\eta(t) \geq 0$ for all $t \in \mathcal{T}$. Moreover, $K_{\mathrm{RISE}} > \overline{D}$ implies $\zeta(t) \geq 0$ for all $t \in \mathcal{T}$. Therefore, $P(t) = \zeta(t) + \eta(t) \geq 0$ for all $t \in \mathcal{T}$. ■

Define the augmented ensemble error state $z_P : \mathcal{T} \to \mathbb{R}^{3n+1}$ as $z_P \triangleq \begin{bmatrix} z^\top & P \end{bmatrix}^\top$. Using (17) and (19), the evolution of z_P is a Filippov solution to the augmented initial value problem

$$\dot{z}_P \in \psi(z_P, t), \quad z_P(t_0) = z_{P,0}, \tag{20}$$

where $z_{P,0} \triangleq \begin{bmatrix} z_0^\top & P(t_0) \end{bmatrix}^\top$ and $\psi : \mathbb{R}^{4n+1} \times \mathbb{R}_{\geq 0} \rightrightarrows \mathbb{R}^{4n+1}$ is the set-valued map

$$\psi(z_P, t) \triangleq \left\{\begin{bmatrix} \xi \\ -\lambda_P P + r_2^\top\left(K_{\mathrm{RISE}}\upsilon_P - D_t\right) \end{bmatrix}\right. : \left.\begin{pmatrix} \xi \in G(z, t), \\ \upsilon_P \in K[\mathrm{sgn}](r_1) \end{pmatrix}\right\}.$$

Since $K[\mathrm{sgn}](r_1)$ is nonempty, compact, convex for every $r_1 \in \mathbb{R}^n$, and since $G(z,t)$ is formed from affine functions of z, D_t, and ν, $G(z,t)$ is a nonempty, compact, convex subset of $\mathbb{R}^{3n}$. Thus, each value of $\psi(z_P, t)$ is a nonempty, compact, convex subset of $\mathbb{R}^{3n+1}$. The graph of $K[\mathrm{sgn}](r_1)$ is closed, $t \mapsto D_t$ is continuous, and all remaining dependencies in G and in $(z_P, \upsilon_P, t) \mapsto -\lambda_P P + r_2^\top\left(K_{\mathrm{RISE}}\upsilon_P - D_t\right)$ are continuous in their corresponding variables, so ψ is upper semicontinuous in (z_P, t). For every bounded cylinder $Z_P \subset \mathbb{R}_{\geq 0} \times \mathbb{R}^{3n+1}$, the quantities e, r_1, r_2 and P are bounded on Z_P, and every $\nu \in \mathrm{SGN}(s)$ satisfies $\|\nu\| \leq \sqrt{n}$. Since $\left|-\lambda_P P + r_2^\top\left(K_{\mathrm{RISE}}\upsilon_P - D_t\right)\right| \leq \lambda_P|P| + \|r_2\|\left(\overline{D} + K_{\mathrm{RISE}}\sqrt{n}\right)$, all the components of $G(z,t)$ are bounded on Z_P. Thus, ψ has nonempty, bounded, closed convex values, is upper semicontinuous, and is locally bounded on bounded cylinders. By Filippov's existence theorem for differential inclusions in [12, Chapter 2, §7, Theorem 1], for each $z_{P,0} \in \mathbb{R}^{4n+1}$, there exists a maximal existence time $T_{\max} \in (t_0, \infty]$ and at least one absolutely continuous Filippov solution $z_P : \mathcal{T} \to \mathbb{R}^{3n+1}$ satisfying $\dot{z}_P \in \psi(z_P, t)$ for almost all $t \in \mathcal{T}$ and $z_P(t_0) = z_{P,0}$. A solution similarly exists for (17).

By studying solutions to (20), sufficient conditions are presented next to ensure completeness and exponential convergence of all Filippov solutions to (17) with initial states z_0 lying in a prescribed compact ball that can be made arbitrarily large.

Theorem 2. *Let $\lambda \in \mathbb{R}_{>0}$ be a desired convergence rate. Let $\Delta \in \mathbb{R}_{>0}$ be a radius for a desired region of attraction for an initial ensemble error $z_0 \in \mathcal{S} \triangleq \left\{\zeta \in \mathbb{R}^{3n} : \|\zeta\| \leq \Delta\right\}$. Select $k_1, k_2, k_3, K_{RISE} \in \mathbb{R}_{>0}$ such that $k_1 \geq 1 + \lambda$, $k_2 > \max\{1+\lambda, 2\lambda\}$, $K_{RISE} \geq \overline{D} + \frac{\overline{\dot{D}}}{k_2 - 2\lambda}$, and $k_3 \geq \lambda + \rho(k_1, k_2, C_\Delta)$, where $C_\Delta \triangleq \sqrt{\Delta^2 + 2C_P\Delta}$, $C_P \triangleq \overline{D} + K_{RISE}\sqrt{n}$, and $\rho(k_1, k_2, s) \triangleq \rho_0^2(k_1, k_2, s) + \rho_0(k_1, k_2, s)$. Suppose that Assumptions 1-6 hold and that the chosen update law and initialization method for $\widehat{\theta}$ satisfy Properties 1-2 for the approximation domain Ω with radius $R_\Omega = \overline{\kappa}$, where $\overline{\kappa}$ is defined according to the C_Δ that follows from the given Δ and selected gains. Then, the controller in 13 guarantees that all Filippov solutions to (17) with $(z_0, t_0) \in \mathcal{S} \times \mathbb{R}_{\geq 0}$ are complete and converge to the origin with the estimate*

$$\|z(t)\| \leq \alpha(\|z_0\|)\,\mathrm{e}^{-\lambda(t-t_0)}, \tag{21}$$

for almost all $t \in \mathbb{R}_{\geq t_0}$, where $\alpha(s) \triangleq \sqrt{s^2 + 2C_P s}$. Consequentially, $z = 0_{3n}$ is uniformly semi-globally asymptotically stable (USAS) for (17). Further, $t \mapsto u(t)$ and $t \mapsto \dot{u}(t)$ belong to the class $\mathcal{L}_\infty(\mathbb{R}_{\geq 0}; \mathbb{R}^n)$ and $\kappa(t) \in \Omega$ for all $t \in \mathbb{R}_{\geq t_0}$.

Proof: Consider any $t_0 \in \mathbb{R}_{\geq 0}$. Select $\lambda_P = 2\lambda$. Since $k_2 > 2\lambda$, the interval $(0, k_2)$ is nonempty. Thus, the condition on K_{RISE} ensures that $P(t) \geq 0$ for all $t \in \mathcal{T}$ by Proposition 1. Consider the candidate Lyapunov function

$$V(z_P) \triangleq \frac{1}{2}z^\top z + P. \tag{22}$$

Since $P(t) \geq 0$ for all $t \in \mathcal{T}$, it follows that $V(z_P(t)) \geq 0$ for all $t \in \mathcal{T}$. Since z_P is absolutely continuous on $\mathcal{T}$ and V of class $\mathcal{C}^1$, $t \mapsto V(z_P(t))$ is absolutely continuous on $\mathcal{T}$. Therefore, the $\dot{V}(t) \triangleq \frac{\mathrm{d}}{\mathrm{d}t}V(z_P(t)) = \left(\nabla V(z_P(t))\right)^\top \dot{z}_P(t)$ for almost all $t \in \mathcal{T}$, where $\nabla V(z_P) = \begin{bmatrix} e^\top & r_1^\top & r_2^\top & 1 \end{bmatrix}^\top$. The derivative of (22) along the trajectories of (20) satisfies

$$\begin{aligned}\dot{V}(t) \leq &- k_1\|e(t)\|^2 - k_2\|r_1(t)\|^2 - k_3\|r_2(t)\|^2 \\ &+ \|e(t)\|\|r_1(t)\| + K_{\mathrm{RISE}}r_2^\top(\nu_2 - \nu_1) \\ &- 2\lambda P(t) + r_2^\top \widetilde{N}_t,\end{aligned} \tag{23}$$

for all $t \in \mathcal{T}$. Since $C_\Delta \geq \Delta$, $\mathcal{D} \supseteq \mathcal{S}$ and $\mathcal{T}$ is nonempty. By [10, Lemma 4], $\nu_2 \underset{\text{a.e.}}{=} \nu_1$. Using this fact, Young's inequality, and the bound on $\left\|\widetilde{N}_t\right\|$, it follows that

$$\begin{aligned}\dot{V}(t) \leq & -(k_1-1)\|e(t)\|^2-(k_2-1)\|r_1(t)\|^2 \\ & -(k_3-\rho(k_1,k_2,C_\Delta))\|r_2(t)\|^2-2\lambda P(t),\end{aligned} \tag{24}$$

for almost all $t \in \mathcal{T}$. Under the gain conditions for k_1, k_2, and k_3 and using (22), it follows that $\dot{V}(t) \leq -2\lambda V(z_P(t))$ for almost all $t \in \mathcal{T}$. Since $V : \mathbb{R}^{3n+1} \to \mathbb{R}$ is of class $\mathcal{C}^0$, [13, Theorem 6.2] implies that

$$V(z_P(t)) \leq V(z_{P,0})\,\mathrm{e}^{-2\lambda(t-t_0)}, \tag{25}$$

for almost all $t \in \mathcal{T}$. Since $\eta(t_0)=0$, it follows that $P(t_0) = \phi(t_0) = K_{\text{RISE}}\|r_1(t_0)\|_1 - r_1^\top(t_0)D_{t_0}$. Thus, $|P(t_0)| \leq C_P\|z_0\|$ and $V(z_{P,0}) \leq \frac{1}{2}\|z_0\|^2 + C_P\|z_0\|$. Since $P(t) \geq 0$, $\frac{1}{2}\|z(t)\|^2 \leq V(z_P(t))$ for all $t \in \mathcal{T}$. Consequentially, (25) implies that

$$\|z(t)\| \leq \beta(\|z_0\|, t-t_0), \tag{26}$$

for almost all $t \in \mathcal{T}$, where $\beta(\|z_0\|, t-t_0) \triangleq \alpha(\|z_0\|)\,\mathrm{e}^{-\lambda(t-t_0)}$. Since $z_0 \in \mathcal{S}$, $\|z_0\| \leq \Delta$. Consider the sublevel set $\Omega_{\mathcal{S}} \triangleq \left\{\zeta \in \mathbb{R}^{3n+1} : V(\zeta) \leq \frac{1}{2}\Delta^2 + C_P\Delta\right\}$. Thus, $z_0 \in \mathcal{S}$ implies that $z_{P,0} \in \Omega_{\mathcal{S}}$. Any $z_P(t) \in \Omega_s$ satisfies $\|z(t)\| \leq C_\Delta$. By (25), it follows that $\Omega_{\mathcal{S}}$ is forward invariant and that all solutions are complete: $\mathcal{T} = \mathbb{R}_{\geq t_0}$. Since β does not depend on z_0 or t_0, the convergence in (26) is uniform. Since β is of class $\mathcal{C}^0$, strictly increasing with respect to $\|z_0\|$, strictly decreasing with respect to $t-t_0$, and Δ can be made arbitrarily large, $z = 0_{3n}$ is USAS for (17) [14, Definition 1].

By (25), $V \in \mathcal{L}_\infty(\mathbb{R}_{\geq 0};\mathbb{R}_{\geq 0})$. Thus, (22), (6), (7), and the definition of z imply that $e, r_1, r_2, \dot{e}, \ddot{e} \in \mathcal{L}_\infty(\mathbb{R}_{\geq 0};\mathbb{R}^n)$ and $P \in \mathcal{L}_\infty(\mathbb{R}_{\geq 0};\mathbb{R}_{\geq 0})$. By Assumption 1, it follows that $q, \dot{q}, \ddot{q} \in \mathcal{L}_\infty(\mathbb{R}_{\geq 0};\mathbb{R}^n)$. By triangle inequality, $\|u\| \leq \overline{u}_{\mathrm{d}} + \|\widetilde{u}\|$. Since $z \in \mathcal{L}_\infty\left(\mathbb{R}_{\geq 0};\mathbb{R}^{3n}\right)$, the bound on $\|\widetilde{u}\|$ implies that $u \in \mathcal{L}_\infty(\mathbb{R}_{\geq 0};\mathbb{R}^n)$. Consequentially, $\kappa \in \mathcal{L}_\infty\left(\mathbb{R}_{\geq 0};\mathbb{R}^{5n}\right)$. It was shown that any trajectory $t \mapsto z(t)$ with $z_0 \in \mathcal{S}$ satisfies $\|z(t)\| \leq C_\Delta$ for all $t \in \mathbb{R}_{\geq t_0}$. By the definition of $\overline{\kappa}$, $\|\kappa(t)\| \leq \overline{\kappa}$ for all $t \in \mathbb{R}_{\geq t_0}$. Thus, selecting $R_\Omega = \overline{\kappa}$ ensures $\kappa(t) \in \Omega$ for all $t \in \mathbb{R}_{\geq t_0}$ and Property 2 always holds. Using this fact, Property 1, and that fact that Φ is of class $\mathcal{C}^0$, EVT guarantees that $\Phi \in \mathcal{L}_\infty(\mathbb{R}_{\geq 0};\mathbb{R}^n)$. Thus, $\dot{u} \in \mathcal{L}_\infty(\mathbb{R}_{\geq 0};\mathbb{R}^n)$ by (12). ∎

VI. Simulations

An integral of time-weighted square-error cost function was used to penalize later tracking errors more than transient ones:

$$J = \int_0^{t_f}\left(q_e \cdot t\|e(t)\|^2 + r_u\|u(t)\|^2\right)dt + W_{\text{fail}}(t_{\text{fail}}-t)^2,$$

where $r_u = 0.2$, $q_e = 1$, $W_{\text{fail}} = 1000$, $t_f \in \mathbb{R}_{>t_0}$ is the total simulation duration, and $t_{\text{fail}} \in (t_0, t_f)$ is time at which the drone escapes the simulated space ($t_f = t_{\text{fail}}$ if it does not escape). The integral was evaluated using the trapezoidal rule.

VII. Experiments

Experimental validation of the developed controller is conducted with acceleration-level control of a ModalAI Sentinel Development Drone with a Mircohard Modem Carrier Board for wireless/RF communication. The vehicle weighs 1.396 kg with its battery, has a payload capacity of 1 kg and a thrust to weight ratio of 2.32. The quadrotor is equipped with the ModalAI VOXL 2 onboard computer and autopilot built on the Qualcomm QRB5165 CPU and running Ubuntu 18.04. The VOXL 2 directly runs PX4 version 1.14 firmware.

The indoor test space is 25 `m` by 12 `m` by 8 `m` and consists of 59 Prime 41 ethernet cameras and a Windows 10 PC running Optitrack Motive. Pose estimates were streamed in real-time at 120 Hz over Nat Net, which were then forward to the quadcopter over an Ubuntu laptop to the quadcopter via the Microhard. Linear velocity estimates were generated with a least-squares estimator running on the mocap pose estimates.

A ROS 2 Humble node running locally on the quadcopter via an Ubuntu 22.04 Docker container encapsulates the controller logic. Three controllers are compared. A baseline RISE controller was

$$u_{\text{BASE}}(t) \triangleq u_{\text{RISE}}(t) + u_{\text{PID}}(t).$$

A RISE controller with a feedforward ResNet ("NN") is

$$\begin{aligned} u_{\text{NN}}(t) &\triangleq u_{\text{RISE}}(t) + u_{\text{PID}}(t) + \Phi\left(Q(t), \widehat{\theta}(t)\right), \\ \dot{\widehat{\theta}}(t) &= \operatorname{proj}_{\mho}\left(\Gamma\left(\left.\frac{\partial\Phi}{\partial\theta}\right|_{(Q(t),\hat{\theta}(t))}\right)^\top r_1(t) - \sigma_{\text{mod}}\widehat{\theta}(t)\right),\end{aligned}$$

where $Q(t) \triangleq \begin{bmatrix} q^\top(t) & \dot{q}^\top(t) & q_d^\top(t) & \dot{q}_d^\top(t)\end{bmatrix}^\top \in \mathbb{R}^{4n}$ and $\sigma_{\text{mod}} \in \mathbb{R}_{>0}$ is the sigma-mod modification gain or forgetting factor, $\Gamma \succ 0$ is a fixed learning rate, and $\mho = \overline{B}_{\overline{\theta}}(0_{\mathrm{p}})$ is the parameter search space. Throughout, $\overline{\theta} = 10^3$ was chosen. Since the desired trajectory is given in a 3D position, $n = 3$ and $q_{\mathrm{d}}(t) = \begin{bmatrix} x_{\mathrm{d}}(t) & y_{\mathrm{d}}(t) & z_{\mathrm{d}}(t)\end{bmatrix}^\top$, which is assumed to be north-east-down (NED) frame convention. The developed controller is called the "integrated RISE neural network" controller ("INN") and is

$$\begin{aligned} u_{\text{INN}}(t) &\triangleq u_{\text{PID}}(t) + u_{\text{RISE}}(t) + \int_{t_0}^{t}\left(\Phi\left(\kappa(\tau), \hat{\theta}(\tau)\right)\right)d\tau, \\ \dot{\widehat{\theta}}(t) &= \operatorname{proj}_{\mho}\left(\Gamma\left(\left.\frac{\partial\Phi}{\partial\theta}\right|_{(\kappa(t),\hat{\theta}(t))}\right)^\top r_1(t) - \sigma_{\text{mod}}\widehat{\theta}(t)\right),\end{aligned}$$

A super-twisting ("ST") controller is

$$\begin{aligned} u_{\text{ST}}(t) = {} & k_{\text{ST},2}\|r_1(t)\|^{1/2}\operatorname{sgn}(r_1(t)) + k_{\text{ST},1}\dot{e}(t) \\ & + k_{\text{ST},3}\int_{t_0}^{t}\operatorname{sgn}(r_1(\tau))\,d\tau,\end{aligned}$$

where $k_{\text{ST},1}, k_{\text{ST},2}, k_{\text{ST},3} \in \mathbb{R}_{>0}$.

All controllers were run at 50 Hz. Weight updates and control inputs were computed by assuming a zero-order hold on $\dot{\widehat{\theta}}$ with a discrete least-squares projection on $\widehat{\theta}$ into $\mho$ denoted as $\operatorname{proj}_{\mho} : \mathbb{R}^{\mathrm{p}} \to \mathbb{R}^{\mathrm{p}}$. Control inputs were fed into PX4 as acceleration set-points. PX4's vertical acceleration was limited to $3.0\ \mathrm{m/s^2}$ and its lateral acceleration was limited to $6.0\ \mathrm{m/s^2}$. Integral control terms were computed with a trapezoidal summation and clamping was used to prevent windup. When either the lateral or vertical control components were saturating, weights were frozen: $\dot{\widehat{\theta}}(t) = 0_{\mathrm{p}}$.

Recall that the desired trajectory must be expressible as the solution to a time-invariant ODE, as in (5). Two desired trajectories were tested. The first trajectory is a 3D figure-eight bounded by amplitudes $A_{\mathrm{x}}, A_{\mathrm{y}}, A_{\mathrm{z}} \in \mathbb{R}_{>0}$ centered at altitude $z_{\mathrm{c}} \in \mathbb{R}_{>0}$, which is the unique solution to

$$\ddot{q}_{\mathrm{d}}(t) = h_1(q_{\mathrm{d}}(t), \dot{q}_{\mathrm{d}}(t)),$$
$$h_1(q_{\mathrm{d}}, \dot{q}_{\mathrm{d}}) \triangleq \begin{bmatrix} -4\omega^2 x_{\mathrm{d}} \dot{\tau}^2(q_d) \\ -\omega^2 y_{\mathrm{d}} \dot{\tau}^2(q_d) \\ -16\omega^2 (z_{\mathrm{d}} - z_{\mathrm{c}}) \dot{\tau}^2(q_d) \end{bmatrix} + \frac{\ddot{\tau}(q_{\mathrm{d}}, \dot{q}_{\mathrm{d}})}{\dot{\tau}(q_{\mathrm{d}})} \dot{q}_{\mathrm{d}},$$
$$\dot{\tau}(q_{\mathrm{d}}) = c\left(1 - \alpha\left(\frac{y_{\mathrm{d}}}{A_{\mathrm{y}}}\right)^2\right), \quad c \triangleq (1-\alpha)^{-1/2},$$
$$\ddot{\tau}(q_{\mathrm{d}}, \dot{q}_{\mathrm{d}}) = 2\alpha c\left(\frac{y_{\mathrm{d}}}{A_{\mathrm{y}}^2}\right) \dot{y}_{\mathrm{d}},$$

with initial conditions $q_{\mathrm{d}}(t_0) = \begin{bmatrix} 0 & 0 & z_{\mathrm{c}} \end{bmatrix}^\top$ m and $\dot{q}_{\mathrm{d}}(t_0) = \sqrt{2}\begin{bmatrix} 0.2\pi & 0.2\pi & \frac{\pi}{15} \end{bmatrix}^\top$ m/s, where $\omega \triangleq \frac{2\pi}{T}$, $T \in \mathbb{R}_{>0}$ is the figure-eight's period, and $\alpha \in [0, 1)$ is a speed variation parameter. Parameters were selected as $T = 30.0$ s, $\alpha = 0.5$, $A_{\mathrm{x}} = 1.5$ m, $A_{\mathrm{y}} = 3.0$ m, $A_{\mathrm{z}} = 0.25$ m, and $z_{\mathrm{c}} = -0.75$ m (NED). The resulting curve achieves desired speeds ranging from 0.4 m/s to 1.3 m/s and a maximum acceleration of 0.5 m/s^2. The second trajectory is a four-petal rose curve constrained to a speed of $V_0 \in \mathbb{R}_{>0}$ at a fixed altitude z_{c}, which is the unique solution to

$$\ddot{q}_{\mathrm{d}}(t) = h_2(q_{\mathrm{d}}(t), \dot{q}_{\mathrm{d}}(t)),$$
$$h_2(q_{\mathrm{d}}(t), \dot{q}_{\mathrm{d}}(t)) \triangleq \begin{bmatrix} \left(\ddot{r}(q_{\mathrm{d}}, \dot{q}_{\mathrm{d}}) - r(q_{\mathrm{d}})\dot{\theta}^2(q_{\mathrm{d}})\right)\cos(\theta(q_{\mathrm{d}})) \\ -\left(r(q_{\mathrm{d}})\ddot{\theta}(q_{\mathrm{d}}) + 2\dot{r}(q_{\mathrm{d}}, \dot{q}_{\mathrm{d}})\dot{\theta}(q_{\mathrm{d}})\right)\sin(\theta(q_{\mathrm{d}})), \\ \left(\ddot{r}(q_{\mathrm{d}}, \dot{q}_{\mathrm{d}}) - r(q_{\mathrm{d}})\dot{\theta}^2(q_{\mathrm{d}})\right)\sin(\theta(q_{\mathrm{d}})) \\ +\left(r(q_{\mathrm{d}})\ddot{\theta}(q_{\mathrm{d}}) + 2\dot{r}(q_{\mathrm{d}}, \dot{q}_{\mathrm{d}})\dot{\theta}(q_{\mathrm{d}})\right)\cos(\theta(q_{\mathrm{d}})), \\ 0 \end{bmatrix},$$
$$r(q_{\mathrm{d}}) \triangleq \sqrt{x_{\mathrm{d}}^2 + y_{\mathrm{d}}^2}, \quad \theta(q_{\mathrm{d}}) \triangleq \mathrm{atan2}(y_{\mathrm{d}}, x_{\mathrm{d}}),$$
$$\dot{\theta}(q_{\mathrm{d}}) = \frac{V_0}{A\sqrt{1 + 3\sin^2(2\theta(q_{\mathrm{d}}))}},$$
$$\ddot{\theta}(q_{\mathrm{d}}) = \frac{-3V_0^2 \sin(4\theta(q_{\mathrm{d}}))}{A^2\left(1 + 3\sin^2(2\theta(q_{\mathrm{d}}))\right)^2},$$

with initial conditions $q_{\mathrm{d}}(t_0) = \begin{bmatrix} 0 & 0 & z_{\mathrm{c}} \end{bmatrix}^\top$ m and $\dot{q}_{\mathrm{d}}(t_0) = \begin{bmatrix} -\frac{\sqrt{2}}{2}V_0 & -\frac{\sqrt{2}}{2}V_0 & 0 \end{bmatrix}^\top$ m/s, where $V_0 = 1.0$ m/s, $A = 2.5$ m, $z_{\mathrm{c}} = -0.5$ m. Although the speed is constant, the curvature induces a maximum acceleration of 2.0 m/s^2.

For the first trajectory, two K-Tool International 42-inch Belt Drive Drum Fans with an air flow capacity of 14800 cubic feet per minute at 537 RPM were placed at $(x_1, y_1) = (-2, 3)$ m and $(x_2, y_2) = (-2, -3)$ m, aligned with the x-axis such that the quadcopter will face a headwind traveling in the direction of the desired trajectory. For the second trajectory, they were moved to $(x_1, y_1) = (3, 0)$ m and $(x_2, y_2) = (0, 3)$ m, respectively, each facing toward the origin.

For each controller and each desired trajectory, four trials are run, each starting from a different initial condition: $(x_0, y_0) \in \{(2.0, 2.0), (-2.0, 2.0), (2.0, -2.0), (-2.0, -2.0)\}$ m.

The update law was chosen...

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